

TS/TM/TL Series GII

Combination with Full Digital Welding Power Source Achieves Stable and High Quality Welding

Torch type selectable to fit your application!

TM series



Separate Type



Through-Arm Type



External Type

TM
1100
1400
1600
1800
2000

Space saving & high payload!

TS-950

TS series



External Type

Through-Arm Type

TS
800
950

Payload
8 kg
TS-800/950

Long-arm & high payload!

TL series



External Type

TL
1800
2000

Payload
TL-1800: **8 kg**
TL-2000: **6 kg**

Manipulator Lineup (as of March 2019)

	TS series		TM series					TL series	
	800	950	1100	1400	1600	1800	2000	1800	2000
Separate	—	—	○	○	○	○	○	—	—
Through-Arm	○	○	○	○	○	○	○	—	—
External	○	○	○	○	—	—	—	○	○
Payload	8 kg		6 kg		4 kg	6 kg		8 kg	6 kg

CO₂/MAG/MIG Robot System (GII)

Combination with Full Digital Welding Power Source Achieves Stable and High Quality Welding

CO₂/MAG/MIG welding robot system selectable to fit your application

Full Digital CO₂/MAG welding machines GZ4 series

SP-MAG

(Super-imposition Control)

TS	TM	TL
800	1100	1800
950	1400	2000
	1600	
	1800	
	2000	



TM-1400GIII (Separate)



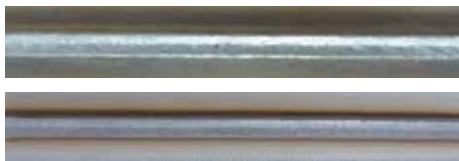
Equipped with TAWERS's SP Control
Praised by Many of Our Customers

SP-MAG benefits:

- Reduced spatter (Reduced removal work)
- Shorter short-circuit cycle suited for high speed welding
- Shorter arc length for good bead appearance

Good bead appearance and low spatter even in high speed welding

Note: Optional parts are necessary to connect GZ4 to robot.



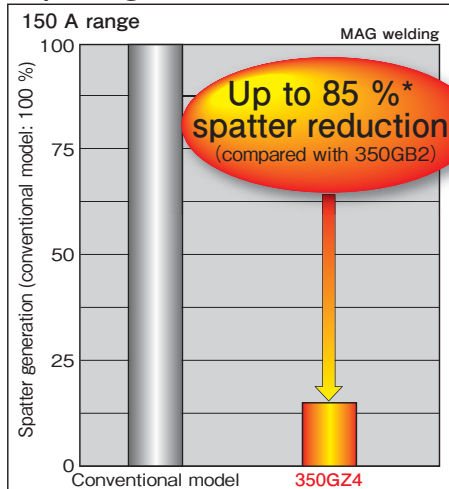
MAG welding (220 A)

- Joint: Fillet • Base metal: Mild steel SPCC (t2.3 mm) • Weld current: 220 A
- Weld speed: 100 cm/min • Wire size: 1.2 mm (YM-50MT)
- Shielding gas: MAG (80 % Ar + 20 % CO₂)

MIG welding (180 A)

- Joint: Fillet • Base metal: SUS308(t1.5 mm) • Weld current: 180 A
- Weld speed: 80 cm/min • Wire size: 1.2 mm (Y308Lsi)
- Shielding gas: MIG (98 % Ar + 2 % O₂)

Spatter generation



*UP to 80% spatter reduction from 500GB2 to 500GZ4 in 250 A range



- Joint: Fillet • Base metal: Mild steel SPCC (t2.3 mm) • Weld current: 150 A
- Weld speed: 50 cm/min • Wire size: 1.2 mm (YM-50MT)
- Shielding gas: MAG (80 % Ar + 20 % CO₂)

Full Digital Controlled Welding Machine

Full Digital

CO₂/MAG/MIG
Welding Machine
Lineup for
High Quality Welds



350GZ4

500GZ4

350VR1TA1

500VR1TA1

400VP1TA1

TIG Robot System (GIII)

High Quality Welding
in Combination with
Full Digital Welding Power Source

Robotic TIG welding system selectable to fit your application

Model Selection Guide

Type	Material	Filler wire dia. (mm)	Welding power source	Robot
Autogenous TIG (no filler)	Stainless steel	—	300BZ3	TS-800 TS-950
	Stainless steel Aluminum	—	300BP4 500BP4	TM-1100 TM-1400 TL-1800
Filler TIG	Stainless steel	1.2	300BZ3	TS-800 TS-950
	Stainless steel Aluminum	1.2	300BP4 500BP4	TM-1100 TM-1400 TL-1800
Rotary filler TIG	Stainless steel	1.2	300BZ3	TL-1800
	Stainless steel Aluminum	1.2	300BP4 500BP4	

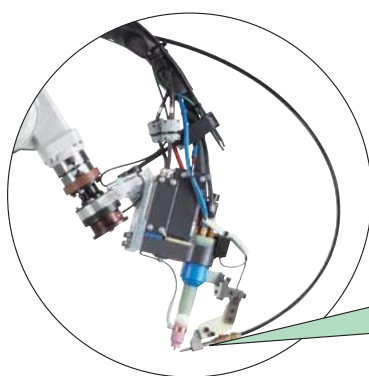
Notes:

- An external axis controller is necessary for rotary filler TIG welding.



Rotary filler TIG Welding Robot System
TL-1800GIII

Features of Rotary TIG Filler Unit



- Optimum welding position
- High precision filler wire feed
- Improved workpiece accessibility

Filler wire tip
position is
adjustable
in all directions.

TIG welding torch lineup



YT-TCT201
• Air-cooled
• 35 % duty cycle at 200 A



YT-TCT401
• Water-cooled
• 60 % duty cycle at 400 A

Full Digital Controlled Welding Machine

Full Digital

TIG Welding Machine Lineup
That Offers High Quality Welding

AC/DC TIG welding machine



300BP4



500BP4

DC TIG welding machine



300BZ3